

Python Best Practices Notes

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1. PEP 8 (Code Style Guide)

- Use snake_case for variables and functions
- Use PascalCase for classes
- Use UPPER_CASE for constants
- Use 4 spaces for indentation
- Keep line length under 88 characters

2. Clean & Readable Code

- Use meaningful variable names
- Keep functions small and focused
- Write readable and simple logic

3. Pythonic Iteration

- Use direct iteration: for item in list
- Avoid range(len()) when unnecessary
- Use enumerate() when index is needed

4. Use Built-in Functions

- sum(), max(), min(), sorted(), len()
- Prefer built-in functions over manual loops

5. List Comprehension

- Use list comprehension for cleaner loops
- Example: [x*x for x in range(5)]

6. Functions Best Practice

- Single Responsibility Principle
- Use type hints
- Write docstrings

7. Exception Handling

- Use try-except properly
- Catch specific exceptions (e.g., ValueError)

8. Logging (Production Ready)

- Use logging instead of print()
- Set proper logging level (INFO, ERROR)

9. Virtual Environment

- Use `python -m venv env`
- Keep project dependencies isolated

10. requirements.txt

- Use `pip freeze > requirements.txt`
- Install with `pip install -r requirements.txt`

11. DRY Principle

- Don't Repeat Yourself
- Reuse code via functions

12. Advanced Practices

- Never hardcode secret keys

- Use .env files
- Follow SOLID principles
- Write modular and maintainable code